Data 102, Fall 2026
Homework 5

Team Effects

1. Different research teams can reach surprisingly different conclusions even when given the same question and the same data.

   To explore the variability associated with analytical choices, this problem will ask you to work through an example data science project. Your goal will be to build an accurate predictive model that estimates diabetes risk using logistic regression.

   Unlike other assignments this semester, *we will collect responses for this problem.* We will use submitted responses to study the variability in results among Data 102 students. Please only submit if you are genuinely interested in the exercise and contributing to the study!

   Participation is optional. You will not lose any course credit if you do not submit. This question is *not* in scope for your final exam. To participate, you must submit a completed project, to Pensive, by 11:59 pm on *Monday, November 23rd.*

   This project will be based on an exercise in Veridical Data Science (VDS) by Bin Yu and Rebecca Barter. The book is available for free, online, at https://vdsbook.com/. All materials and data are linked from the VDS book at https://github.com/Yu-Group/vds-book-supplementary/tree/main. Specific links are available in the problem text.

   (a) Please read Chapter 1.3: "Setting Up Your Data Science Project". This chapter includes good suggestions for organizing your work in a format that will make it easier for you to check the stability of your model, to quickly test multiple approaches, and for others to reproduce your work.

   (b) Then, complete parts (a - e) of Problem 27 in Chapter 11, with the following modifications:

      1. Only fit a logistic regression model.
      2. Test at least two different approaches.[1]
      3. Train on a subset of *at most* 9 of the 11 predictive features isolated in the simplified data set. You may use fewer features, and are free to use any procedure you like to select the features you adopt, including testing multiple models. You must explain, in your submission, how you selected the features you use. You

[1] vary the stage of your approach where you believe there is the most latitude for other approaches and that you expect to have the largest impact on predictions.

may reference the EDA performed in part (d) to justify your choice. Select the features you believe will produce the most accurate predictive model.

Prepare, for submission:

1. A zip file containing all of your code, and a ReadMe file that another researcher could use to navigate your repository.
2. A python notebook that presents your analysis. You may define functions outside the notebook that the notebook references. All functions called must be available in your repository. Make sure your notebook calls all necessary packages so another student could run your notebook without modification.
   Your notebook should justify the choices you made when constructing your predictive model. Write so that a reader could reproduce your results and your reasoning.
3. Select, among the approaches attempted, the model you would expect to have the smallest overall error rate (false positives + false negatives) if deployed on a population with the same prevalence of diabetes as the population provided in the full NHANES dataset. Report:
   (a) Estimated false positive and false negative rates using your validation set.
   (b) Interval estimates for the coefficients of your regression model in the original units provided in the data set.[2] Explain the source of uncertainty represented by your estimates.[3]
   (c) A ranking of your selected features from those you think are most important for prediction to those you think are least important. Explain your procedure for evaluating the importance of features.[4]

[2] If you standardized your data before training, convert back so we can compare across submissions.

[3] e.g. variability over plausible hypothetical data sets via bootstrapping the training data, posterior uncertainty given a selected prior model for the regression coefficients, variability with respect to data cleaning or feature selection decisions.

[4] e.g. size of coefficient after standardizing each predictive feature by subtracting off its mean and scaling by its standard deviation over the training set, reduction in predictive accuracy of your model if you drop the feature from your feature list.

Backpropagation with Binary Cross Entropy (BCE) Loss

2. Similar to Discussion 9, consider a two-layer neural network that computes a real-valued function of the form $f_{w_1,w_2}(x) = \sigma(w_2\sigma(w_1 x))$ where $x, w_1, w_2 \in \mathbb{R}$, and σ is the sigmoid function given by $\sigma(x) = \frac{1}{1+e^{-x}}$.

   The neural network f_{w_1,w_2} can be trained to predict a real-valued output given an m-dimensional input (a regression problem). Given a data set of n input-output pairs, $\{(x_i, y_i)\}_{i=1}^n$, a common way of training a neural network to perform this task is to find the parameter values w_1 and w_2 that minimize the Binary Cross Entropy (BCE) loss over the data set:

$$\arg\min_{w_1,w_2} \sum_{i=1}^{n} -y_i \log(f_{w_1,w_2}(x_i)) - (1-y_i)\log(1 - f_{w_1,w_2}(x_i)) \tag{1}$$

   This can be more explicitly written by replacing $f_{w_1,w_2}(x)$ with its equivalent expression in this problem, $w_2\sigma(w_1 x)$.

$$\arg\min_{w_1,w_2} \sum_{i=1}^{n} -y_i \log(\sigma(w_2\sigma(w_1 x_i))) - (1-y_i)\log(1 - \sigma(w_2\sigma(w_1 x_i))) \tag{2}$$

   For simplicity, we will focus on the partial derivatives of the BCE loss evaluated on a single data point (x, y):

$$\mathcal{L}(w_1, w_2) = -y \log(\sigma(w_2\sigma(w_1 x))) - (1-y)\log(1 - \sigma(w_2\sigma(w_1 x))) \tag{3}$$

   We can also define the intermediate quantities as follows:

$$\begin{aligned} z_1 &= w_1 x & z_2 &= \sigma(z_1) & z_3 &= w_2 z_2 & \hat{y} &= \sigma(z_3) \\ z_4 &= \log(\hat{y}) & z_5 &= \log(1-\hat{y}) & z_6 &= -y z_4 & z_7 &= -(1-y) z_5 \end{aligned}$$

   (a) *Drawing a Computation Graph*

   For the loss function defined above in equation (5), draw four separate computation graphs. Label all the intermediate quantities $z_1, z_2, z_3, \hat{y}, z_4, z_5, z_6, z_7$ in the graph. First, make a computation graph for $\hat{y}$ as a function of x, w_1, and w_2. Next, make two separate computation graphs: one for z_6 and one for z_7, both as functions of y and $\hat{y}$. Finally, make a fourth graph for $\mathcal{L}(w_1, w_2)$ as a function of z_6 and z_7. Using four computation graphs will make taking the derivatives later in the problem easier since having only one computation graph is quite messy.

   (b) *(Sigmoid Function Derivative*

   Show that $\frac{d}{dx}\sigma(x) = \sigma(x)(1-\sigma(x))$.

   (c) *Updating w_2*

   Using the chain rule, write down an expression for the derivative of the loss function with respect to w_2: $\frac{\partial \mathcal{L}(w_1,w_2)}{\partial w_2}$. Use intermediate quantities from the forward pass $(z_1, \ldots, z_7)$ listed above wherever possible, since these have already been computed after the forward pass.

(i) Write $\frac{\partial \mathcal{L}(w_1,w_2)}{\partial w_2}$ in terms of $\frac{\partial \mathcal{L}(w_1,w_2)}{\partial z_7}$, $\frac{\partial \mathcal{L}(w_1,w_2)}{\partial z_6}$, $\frac{\partial z_7}{\partial w_2}$, and $\frac{\partial z_6}{\partial w_2}$ using the chain rule.

(ii) Compute $\frac{\partial z_7}{\partial w_2}$ and $\frac{\partial z_6}{\partial w_2}$ as a function of x, y, w_1, w_2, and the intermediate quantities from the forward pass $z_1, \ldots, z_7$. To receive full credit, your answers should use the intermediate quantities as much as possible, and avoid redundant computation.
Hint: It may be useful to also expand $\frac{\partial z_7}{\partial w_2}$ and $\frac{\partial z_6}{\partial w_2}$ using the chain rule first.
Hint: You may use the result from Part B without proof.

(iii) Use the previous two parts to compute $\frac{\partial \mathcal{L}(w_1,w_2)}{\partial w_2}$ as a function of x, y, w_1, w_2, and the intermediate quantities from the forward pass $z_1, \ldots, z_7$. To receive full credit, your answers should use the intermediate quantities z_i as much as possible, and avoid redundant computation.

(d) *Updating w_1*

Using the chain rule, write down an expression for $\frac{\partial \mathcal{L}(w_1,w_2)}{\partial w_1}$. Once again, use the intermediate quantities from the forward pass wherever possible.

(i) Write $\frac{\partial \mathcal{L}(w_1,w_2)}{\partial w_1}$ in terms of $\frac{\partial \mathcal{L}(w_1,w_2)}{\partial z_7}$ $\frac{\partial \mathcal{L}(w_1,w_2)}{\partial z_6}$, $\frac{\partial z_7}{\partial w_1}$, and $\frac{\partial z_6}{\partial w_1}$ using the chain rule.

(ii) Compute $\frac{\partial z_7}{\partial w_1}$ and $\frac{\partial z_6}{\partial w_1}$ as a function of x, y, w_1, w_2, and the intermediate quantities from the forward pass $z_1, \ldots, z_7$. To receive full credit, your answers should use the intermediate quantities z_i as much as possible, and avoid redundant computation.
Hint: It may be useful to also expand $\frac{\partial z_7}{\partial w_1}$ and $\frac{\partial z_6}{\partial w_1}$ using the chain rule first.
Hint: Some of the quantities you computed in Part C may be useful.
Hint: After solving for one quantity, the other needs no new derivatives (i.e. if you solve for $\frac{\partial z_7}{\partial w_1}$ first, then $\frac{\partial z_6}{\partial w_1}$ needs no new derivatives and vice versa).

(iii) Use the previous two parts to compute $\frac{\partial \mathcal{L}(w_1,w_2)}{\partial w_1}$ as a function of x, y, w_1, w_2, and the intermediate quantities from the forward pass $z_1, \ldots, z_7$. To receive full credit, your answers should use the intermediate quantities z_i as much as possible, and avoid redundant computation.

(e) *Biases*

Most real networks have free weights and biases. Suppose that we modified the network to include biases in the linear combinations:

$$f_{w_1,b_1,w_2,b_2}(x) = \sigma(w_2\sigma(w_1x + b_1) + b_2).$$

Find $\frac{\partial \mathcal{L}(w_1,b_1,w_2,b_2)}{\partial b_1}$. Reuse as much work as possible.

(f) *Usefulness of Intermediate Values*

Why does saving and reusing the intermediate terms $z_1, \ldots, z_7, \hat{y}$ make computing $\frac{\partial \mathcal{L}(w_1,w_2)}{\partial w_2}$ and $\frac{\partial \mathcal{L}(w_1,w_2)}{\partial w_1}$ more efficient?
Hint: Think about how much more algebra you would have needed to do if all answers above had to be in terms of w_1, w_2, x.